Monitoring and Data Processing Architecture using the FIWARE Platform for a Renewable Energy Systems

Washington Velasquez, Luis Tobar-Andrade and Ivan Cedeno-Campoverde
Escuela Superior Politécnica del Litoral, ESPOL, FIEC, P.O. Box 09-01-5863, Guayaquil, Ecuador
Email: (wavelasq, ljtobar, ilcedeno)@espol.edu.ec

2021 IEEE 11th Annual Computing and Communication Workshop and Conference (CCWC) | 978-1-6654-1490-6/21/$31.00 ©2021 IEEE | DOI: 10.1109/CCWC51732.2021.9376026

***Abstract*—This paper states the model of a computational architecture using the FIWARE platform to be incorporated into renewable energy systems to monitor variables such as power, current, temperature, and more, allowing the evaluation and recognition of patterns that generate notifications in case of an emergency. The results indicate that the architecture is easily adaptable to different environments due to the FIWARE-Orion and FIWARE-Draco components, which allow the creation of dynamic data communication channels providing a reliable, powerful, and adaptable tool.**

***Index Terms*—Architecture, Energy, FIWARE, Monitoring, System**

I. Introduction

Electric energy is a basic and essential need in the daily activities of human beings [1]. However, the different ways of energy production have generated a serious impact on the environment. Ecuador's energy-producing companies explore alternatives in elaborating the resource in a cleaner and more sustainable way to conserve natural resources at risk [2]. This energy unsustainability generated by the limited condition of resources has led man to seek a less polluting possibility of renewable use, such as solar energy. Photovoltaic solar panels generate endless energy helping sustainable development. Likewise, it encourages local employment development, being a system specifically appropriate for rural or isolated areas, where the power line does not supply, or its installation is relatively expensive.

In Ecuador, some companies are dedicated to providing renewable energy alternatives to save on energy consumption costs [3]. A large part of this photovoltaic source equipment does not have a monitoring system included. Over time, they tend to present different problems caused by the continuous exposure of some natural factors such as UV radiation, climatic conditions, and temperature. These problems affect the solar panels' performance since there is no adequate control of the system's operation, generating a decrease in energy production. It is necessary to have an effective monitoring system to measure the variables consistently, thus determining in real-time the performance of the photovoltaic source system, consistently providing this information [4]. The importance of this proposal is to introduce a system that performs real-time monitoring of renewable energies to keep track of energy consumption and the savings generated each time solar panels are used, allowing the incorporation of existing systems that includes concepts such as portability, adaptability, and monitoring without human supervision. In this context, it is desired to implement an integrated monitoring structure for photovoltaic solar panels with a reliable communication model that allows real-time processing of variables such as voltage, current, power, and temperature, allowing to observe these variables' performance to predict behavior by sending alerts if an incident occurs, such as an unexpectedly high voltage or current electrical in order to provide an open-source computational tool.

The rest of the document is divided into five sections. Section II describes state of the art, emphasizing the FIWARE platform, and related works. Section III presents the proposed architecture explaining each of the layers and the interaction with the different technologies. Section IV shows the preliminary results and the tests performed for the measurement of the variables of interest. Finally, the conclusions and future works based on the results obtained are presented.

II. State of the Art

In the market, several methods allow monitoring of renewable energy systems; these generally use technologies and paid software that makes these systems more expensive, e.g., in [5], a SCADA system was made to monitor solar panels with batteries where they develop a design prototype of a SCADA system to monitor and automate photovoltaic solar panels. In [6], the implementation of a monitoring system for the capture and measurement in real-time of a photovoltaic system was made through LabVIEW and Arduino where they develop a control system to be able to observe from experimental data the efficiency curve and the characteristic IV curve (current vs. voltage) through a photovoltaic panel. The development of this aforementioned technology is analogous due to the use of Arduino for the acquisition of data but not from LabVIEW, and in [7] a remote monitoring system for photovoltaic systems was carried out. They show the development of a platform to monitor isolated photovoltaic systems in Ecuador to promote a progressive solution using free software. However, this paper's proposed architecture goes beyond monitoring variables in these systems but rather trying to make these systems intelligent through open-source software tools.

A. FIWARE Open-source Platform

The platform offered by the FIWARE foundation[1] allows the creation of intelligent applications in different environments [8]. The foundation provides open-source tools that developers can use to convert or generate innovative proposals for daily life problems. The components used in the design are:

- **FIWARE-Orion**: (Orion Context Broker - OCB), It allows you to manage the entire lifecycle of information, including updates, queries, registrations, and subscriptions. It is an NGSIv2 server implementation to manage context information and its availability. This component allows the rapid integration of different applications; through the communication model (publish-subscriber), you can send context-based information to entities/systems that need data for their operation.
- **FIWARE-Draco**: This component is based on (Apache Nifi), which allows creating communication channels that can be redirected to multiple applications, e.g., a database, logs, streaming processing, and more. FIWARE-Draco takes care of receiving notifications from FiWARE-Orion to redirect data through architecture components.

B. Related works

The FIWARE platform can be used for the rapid creation of IoT solutions; it allows easy coupling with other media through Context-Broker (Orion), e.g., in [9] a scheme is described where FIWARE is used as a nucleus for the creation of a computer management system. This design uses several components of the platform to manage the computer resources within a university campus or in [10]; it is used in a cloud-based environment that allows the rapid administration of agricultural processes. On the other hand, if we focus on electrical energy management systems, the proposal described in [11] presents a configuration of IoT environments using the FIWARE platform to integrate existing automation networks, linking OpenMUC with BACnet functionalities connected through MQTT. The authors showed the information through a web platform where they use a basic control circuit to integrate the most advanced energy management systems in buildings. Likewise, in [12], the demand for electrical energy in various buildings is analyzed; The authors present the term IoT Energy Platform (IoTEP), providing a tool that collects data from various sensors located within buildings; they indicate that the results obtained were better than expected based on the case studies. It can be seen that electrical energy systems using the FIWARE platform are various and applied in different areas of IoT applications [13]. However, the architecture presented in this paper goes beyond the administration of electrical energy use but tries to use other components that are easily coupled to the FIWARE platform to generate predictions based on events and thus avoid emergencies.

[1] https://www.fiware.org/foundation/

III. Architecture

This section describes the stages, the interaction of the FIWARE components with other open-source modules, and the different functionalities that they perform (Fig. 1).

A. Data Acquisition

Data acquisition is based on capturing signals from the real world, in order to be able to produce data which can be manipulated through an electronic device. These signals cannot be received directly to the electronic device, they have to be transformed into digital signals. In this stage, the objective is to establish communication between the photovoltaic source, which is the means to produce renewable energy obtained directly from solar radiation. The Arduino mega was used as a controller, which is an electronic board with a development environment that is based on free, adaptable and easy-to-use hardware and software, which will help us to extract all the data.

B. Communication (Publish-Subscriber)

The data obtained in the acquisition stage is sent to an information manager. FIWARE-Orion technology was used, which allows managing the entire life cycle of context information; through this technology, the elements (variables) of the entities were created to publish them (producers), so that the information published is available to other entities (consumers) [14]. FIWARE-Draco was used to manage and distribute data, and a subscription was made to Orion to be receiving any notification through events when an inconvenience arises.

The core of communication is the entities. They use an NGSIv2 data model that contains the FIWARE architecture. An entity refers to a thing. The context attributes are characteristics that detail the entities, e.g., power, battery, temperature, energy, and current, and the metadata is pieces of information in charge of detailing some extra properties. Fig. 2 represents the producers (context information), which generates information from the data that is in permanent variation, these values of the variables are found in the entities that were previously established in the OCB so that manage all the context, as well as the consumer of this context information, which will be subscribed to the data of the entities that are established in the OCB to receive all the data updates.

C. Storage (Docker Containers)

Storage within the system represents a crucial stage due to the large variety of data expected to be collected per second. In addition, FIWARE components use specific databases to handle persistence and other features such as logs. The data storage was done through Docker, where a container was created for each database in the system. Two databases were used:

- MongoDB - FIWARE-Orion uses this database to store the entities' current state, but it does not store historical

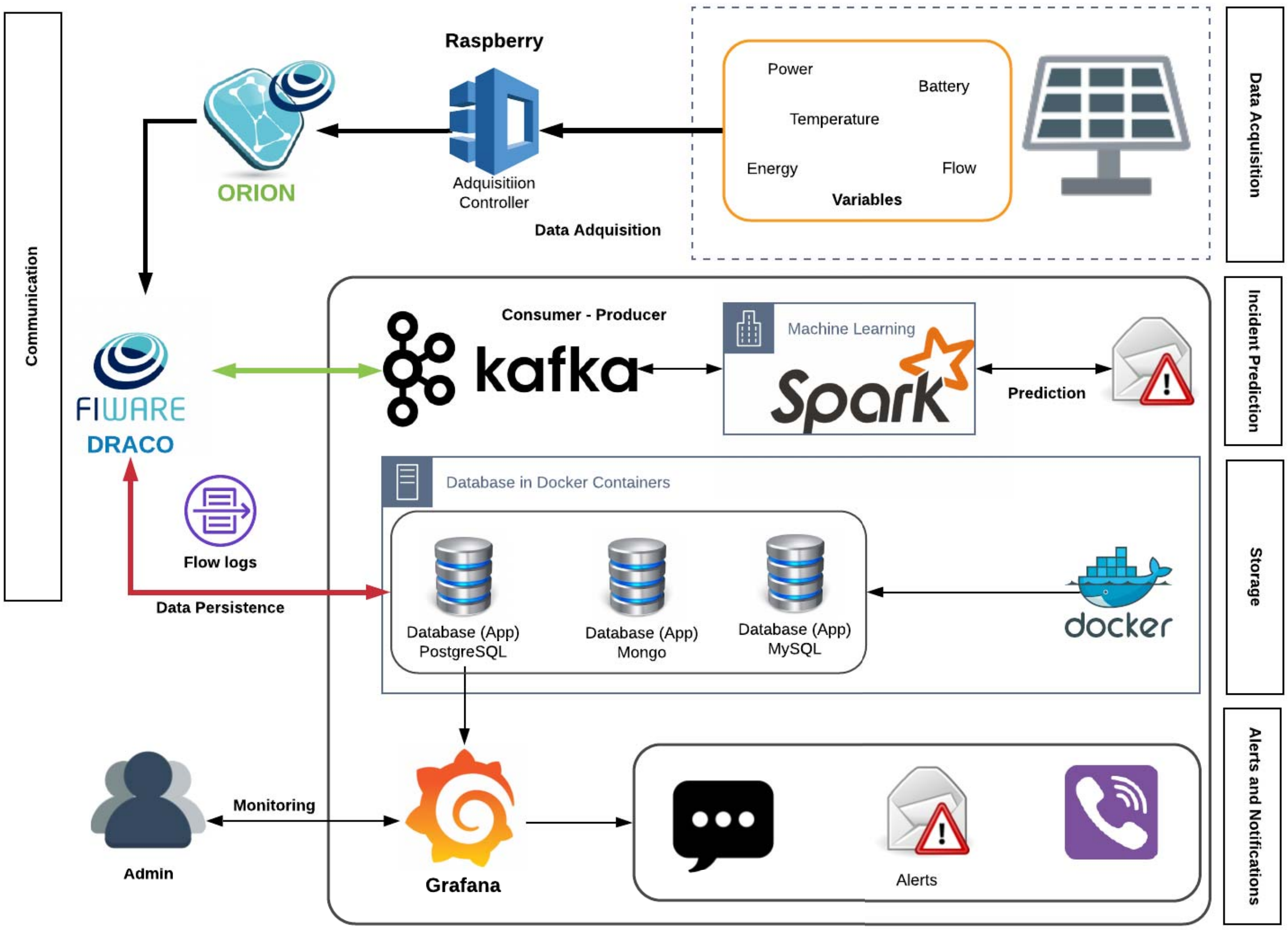

Fig. 1. Architecture proposal for the monitoring and data processing system.

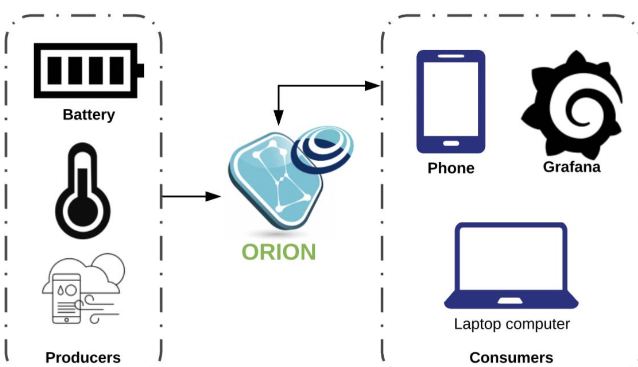

Fig. 2. Subscription to entities (Fiware-Orion).

changes. The information as a history is stored in PostgreSQL.

- PostgreSQL - This database was used to store the system information and the metric values (power, current, temperature), and thus be able to be visualized through Grafana.

D. Incident Prediction

Prediction models are a set of mathematical and programming techniques to establish the possibility of future events based on historical data analysis. This is currently achieved through programs that use automatic learning algorithms, which is the progressive development of finding patterns in the data itself and, once they are found, using that pattern to make new predictions. Apache Spark [15] is an option that provides these features to work efficiently concerning planning, distribution, and monitoring in a cluster (Apache Kafka).

At this stage, as a future implementation, Apache Kafka can be used, which is a platform that manages a large amount of data flow and distributes it to the different users, to maintain the transmission of information through its pub-sub model, which consists of a subscriber expressing his interest in a specific topic or event and then he is notified about the event that occurred [16]. These events are held by publishers. FIWARE-Draco, through its web interface, allows a distribution to other used software where one of those is Apache Kafka, in which the producer is established to publish the topic with the data redirecting to Apache Spark, which is a system that allows processing in time real data, where you subscribe to the same topic to receive and process the data, in this way in the future, using the data that is stored in the database, automatic predictive analysis of how long it will be filled could be performed the batteries, having the data history (dataset) previously and then passing to a machine language algorithm to analyze the data.

E. Alerts and Notifications

Monitoring is one of the fundamental stages because, without this component, it is not feasible to check if the variables that make up the system are operating correctly [17]. The development of the interface is not focused on an application as such, but rather on the use of free software called Grafana [18], a tool to represent time series data. From the data stored in the PostgreSQL database, the variables of interest were monitored through graphs and histograms displayed on the web-platform. Finally, alerts were configured to users establishing thresholds as a reference for each variable.

IV. Analysis of Results

In this section, the results obtained from the system's implementation are detailed to know the real-time status of the variables using different scenarios.

A. Laboratory environment - Equipment daily use.

In Fig. 3, a conventional hairdryer is connected to the monitoring system that will measure its general performance and visualize the energy and the cost it generates from the moment it was turned on, varying its different levels of speed.

Fig. 3. Tests with a hairdryer

In Fig. 4, the levels of the cost generated by the hairdryer are observed and how it is varying over time, thus keeping an exhaustive control of what is being generated in costs to generate Long-term economic savings; using this electronic device momentarily will depend on whether or not it generates a large energy consumption within the house. These reflected values correspond to half an hour of constant use, corresponding to the value kW/h in Ecuador.

Fig. 4. System monitoring

B. Laboratory environment - Solar panels

A solar panel system developed in a scale model that generates energy with some LEDs was monitored, as shown in Fig. 5. Likewise, it can be viewed in the monitoring system (Fig. 4) in real-time as is the general performance of the photovoltaic source, the level of charge that the batteries are, how much energy consumption is flanking towards the homes, and how much It is the money-saving when using a solar panel system concerning the conventional one.

Fig. 5. Model with a solar panel

The battery charge storing energy through the solar panel is 4V, enough to turn on the LEDs that come to represent the light bulbs of a house powered by this type of renewable source. The economic savings generated by this solar panel system is low since this is a scale model; therefore, it does not generate significant savings. This value is better appreciated if a house that has a larger solar panel is monitored.

C. Data Transmission

The transmission of data does not cause significant traffic on the network; this is because the communication model implemented by Fiware-Draco and Orion is based on events that are represented as data, which means that if it does not generate the event, the packet is not sent. To generate the event and send the packet, there must be an update of one of the variables' values, maintaining the data's persistence, and avoiding flooding the network with unnecessary traffic. This is the cost of hiring of bandwidth generates a high economic saving.

To analyze the time between the data sent, a sample of one hour was taken, obtaining; as a result, a total of 3000 readings of the packages, where each one contains information about the data, resulting in the graph of Fig. 6. Taking a total value at the end of the sampling time of 18.2 Mb/s, which gives us a referential value of how much the network will be consuming during that time. Besides, From this value, it is possible to estimate how much bandwidth will be needed to transport this data without any problem maintaining a stable and fluid connection in such a way that the monitoring system constantly works to optimize costs and resources.

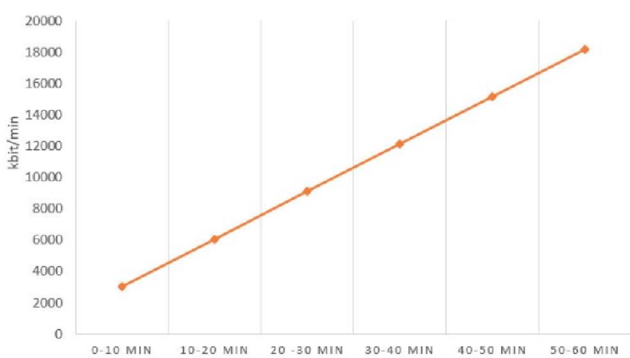

Fig. 6. Data transfer (transmission) over the network

V. Conclusions

The monitoring system's architecture in conjunction with the applied technologies facilitates that the project can be developed and carried out easily and efficiently through FIWARE-DRACO for its implementation, incorporation, and connection with all other services. You can visualize and raise awareness about how much energy is being used irrationally through the monitoring system, and through the communication model based on implemented events, network resources are saved since the data sent is processed by FIWARE-Orion.

The tests that were carried out allow us to verify that the services provided by FIWARE-DRACO accommodate and provide simplicity when working in conjunction with other storage technologies. Taking into account that the database that is chosen as storage of the data is indifferent. The implementation enables it to be established in other environments since the developed software can engage different applications.

Acknowledgment

The authors would like to thank to the Escuela Superior Politécnica del Litoral (ESPOL) for financing the continuity of this study.

References

[1] J. C. Schallenberg Rodríguez, G. Piernavieja Izquierdo, C. Hernández Rodríguez, P. Unamunzaga Falcón *et al.*, "Energías renovables y eficiencia energética," 2008.

[2] R. Gomelsky, "Evaluación rápida y análisis de brechas en el sector energético ((rgt1881) ecuador)," 2013.

[3] J. Guamán-Molina, C. Vargas-Guevara, R. Nogales-Portero, D. Guevara-Aulestia, M. García-Carrillo, and A. Ríos-Villacorta, "Solar manager: plataforma cloud de adquisición, tratamiento y visualización de información de sistemas fotovoltaicos aislados," *Ingenius. Revista de Ciencia y Tecnología*, no. 15, pp. 5–16, 2016.

[4] L. Zhao, J.-l. Zhang, and R.-b. Liang, "Development of an energy monitoring system for large public buildings," *Energy and Buildings*, vol. 66, pp. 41–48, 2013.

[5] A. Salazar, Y. Garzón, G. Zapata, A. Castaño, J. Rincón, and R. García, "Diseño de un scada para la monitorización de una solución solar fotovoltaica con baterías."

[6] B. Vargas, J. Pablo, A. Navia *et al.*, "Desarrollo de un sistema de control para la captura y medición experimental de la eficiencia y curva característica iv en tiempo real de un sistema fotovoltaico utilizando labview® y arduino," *Investigación & Desarrollo*, vol. 1, no. 15, pp. 49–64, 2015.

[7] S. Manzano, R. Peña-Ortiz, D. Guevara, A. Ríos *et al.*, "Plataforma cloud para monitoreo remoto de sistemas fotovoltaicos aislados en el ecuador," 2014.

[8] J. Guth, U. Breitenbücher, M. Falkenthal, F. Leymann, and L. Reinfurt, "Comparison of iot platform architectures: A field study based on a reference architecture," in *2016 Cloudification of the Internet of Things (CIoT)*. IEEE, 2016, pp. 1–6.

[9] W. Velasquez and M. Filian-Gomez, "Communication network model for a computer management and control system implemented using fiware platform: Case study," *IEEE Latin America Transactions*.

[10] J. López-Riquelme, N. Pavón-Pulido, H. Navarro-Hellín, F. Soto-Valles, and R. Torres-Sánchez, "A software architecture based on fiware cloud for precision agriculture," *Agricultural water management*, vol. 183, pp. 123–135, 2017.

[11] T. Storek, J. Lohmöller, A. Kümpel, M. Baranski, and D. Müller, "Application of the open-source cloud platform fiware for future building energy management systems," in *Journal of Physics: Conference Series*, vol. 1343, no. 1. IOP Publishing, 2019, p. 012063.

[12] F. Terroso-Saenz, A. González-Vidal, A. P. Ramallo-González, and A. F. Skarmeta, "An open iot platform for the management and analysis of energy data," *Future Generation Computer Systems*, vol. 92, pp. 1066–1079, 2019.

[13] L. Barreto, A. Celesti, M. Villari, M. Fazio, and A. Puliafito, "Identity management in iot clouds: A fiware case of study," in *2015 IEEE Conference on Communications and Network Security (CNS)*. IEEE, 2015, pp. 680–684.

[14] T. Zahariadis, A. Papadakis, F. Alvarez, J. Gonzalez, F. Lopez, F. Facca, and Y. Al-Hazmi, "Fiware lab: managing resources and services in a cloud federation supporting future internet applications," in *2014 IEEE/ACM 7th International Conference on Utility and Cloud Computing*. IEEE, 2014, pp. 792–799.

[15] X. Meng, J. Bradley, B. Yavuz, E. Sparks, S. Venkataraman, D. Liu, J. Freeman, D. Tsai, M. Amde, S. Owen *et al.*, "Mllib: Machine learning in apache spark," *The Journal of Machine Learning Research*, vol. 17, no. 1, pp. 1235–1241, 2016.

[16] N. Garg, *Apache Kafka*. Packt Publishing Ltd, 2013.

[17] K. Chooruang and K. Meekul, "Design of an iot energy monitoring system," in *2018 16th International Conference on ICT and Knowledge Engineering (ICT&KE)*. IEEE, 2018, pp. 1–4.

[18] A. Aimar, S. Belov, L. Magnoni, P. Saiz, B. Garrido Bear, A. Aguado Corman, P. Andrade, M. Georgiou, E. Karavakis, R. Rama Ballesteros *et al.*, "Unified monitoring architecture for it and grid services," in *J. Phys. Conf. Ser.*, vol. 898, 2017, p. 092033.